

Article

Development and Optimization of Fine-Pitch RDL for RDL Interposer and Embedded Bridge Die Interposer Fabrication Using Fan-Out Wafer-Level Packaging Technology

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Abstract

Fine-pitch redistribution layers (RDLs) are key enabling technologies for fan-out wafer-level packaging (FOWLP)-based interposers used in chiplet and high-bandwidth memory (HBM) integration. In this study, a CAR-based photolithography process optimized for fine-pitch RDL fabrication was evaluated to realize 2 μm /2 μm line/space (L/S) RDL structures in an FOWLP environment. Key lithographic parameters, including exposure energy, focus offset, and thermal processing conditions, were systematically optimized to establish a stable and reproducible process window. Cross-sectional analysis confirmed the structural integrity of the electroplated RDL features formed under the optimized conditions. To assess functional feasibility, channel-level electrical simulations were performed using JEDEC-defined HBM3 signal assignments. Simulated eye diagrams indicate that the fabricated fine-pitch RDL interconnects are capable of supporting HBM3-class signal transmission with a moderate level of signal integrity. The presence of jitter and noise suggests that further optimization of RDL transmission line impedance is required. Rather than presenting a fully optimized interposer solution, this work provides an engineering-level assessment of lithographic and process constraints associated with implementing 2 μm class RDLs in FOWLP-based interposers, offering practical insight into fine-pitch RDL process window definition for advanced packaging applications. This work uniquely combines systematic CAR-based lithography optimization with cross-sectional structural validation and HBM3-class channel-level simulations to define a practical process window for 2 μm /2 μm RDLs in an FOWLP environment.

Keywords: FOWLP (Fan-Out Wafer Level Package); interposer; chiplet; advanced packaging; fine pitch RDL



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1. Introduction

The continued drive for higher integration, bandwidth, and energy efficiency has positioned advanced packaging as a key enabler of system-level performance. Among

various approaches, Fan-Out Wafer-Level Packaging (FOWLP) has gained prominence by eliminating traditional substrates and forming redistribution layers (RDLs) on mold or polymer dielectrics, thereby providing high I/O density, shortened interconnects, and compact form factors [1–3]. This direction aligns with the broader transition toward heterogeneous integration and chiplet-based architectures, where manufacturable, fine-pitch interconnects with low parasitics are required for AI and high-performance computing (HPC) applications [1,4–7].

Despite this progress, reliably realizing fine-pitch RDLs at and below the 5 μm regime over non-planar topographies and relatively thick dielectric stacks typical of FOWLP and panel-level processes remains challenging. Die-to-die interconnections in 2.5D interposers require patterns of 2 μm /2 μm or below line and space [8–10]. Conventional lithography flows are constrained by depth-of-focus limitations, topography-induced image distortion, and variability in thick photoresist processing, which can limit pattern fidelity, yield, and scalability [2,11]. To mitigate these issues, multiple complementary strategies have been investigated. These include material innovations such as chemically amplified positive photoresists (CAR) and tailored underlayers to manage acid generation and diffusion and to reduce line-edge roughness (LER) [11,12]; optical and process advances including high-NA stepper imaging, improved focus/leveling and topography compensation, and enhanced inspection/metrology to tighten CD control across die and panel formats [3]; adoption of dry-film polymer dielectrics and panel-scale lithography to improve coating uniformity and enable high-aspect-ratio trench RDLs on large substrates [3,13]; and interconnect architecture optimization using fine-pitch multi-metal RDL stacks co-optimized for chiplet accelerators and 2.5D/3D integration [1,4]. These efforts have demonstrated the feasibility of 2 μm class line/space (L/S) features and, in some panel processes, sub-micrometer trench structures [1,3,10]. Nevertheless, consistent pattern fidelity, adequate process latitude, and material compatibility across sub-5 μm RDLs require further improvement for high-volume manufacturing [2,8]. Beyond conventional FOWLP and panel-level processes, several recent works have focused on RDL-first FOWLP interposers and large RDL interposer packages to support HBM and chiplet integration. These studies demonstrate that fine-pitch multi-layer RDL stacks can achieve the interconnect density required for high-bandwidth memory and logic-to-logic links, but also highlight challenges related to warpage, dielectric reliability, and alignment over large areas [1,4,5,7].

From the lithography standpoint, most prior work has either focused on demonstrating sub-micrometer feature capability using advanced exposure tools and optimized CAR formulations or on improving coating uniformity and line-edge roughness across large substrates [3,11,13]. However, there is comparatively less emphasis on quantitatively defining a robust process window for 2 μm /2 μm RDL fabrication under realistic FOWLP conditions, where thick dielectric stacks, non-planar topography, and process variability can significantly impact pattern collapse, CD control, and downstream electroplating behavior.

Motivated by these gaps, the present work aims to bridge the divide between high-resolution lithography demonstrations and manufacturable process integration by systematically mapping the lithographic process window for 2 μm /2 μm CAR-based RDLs in an FOWLP environment and by directly correlating the resulting RDL structures with HBM3-class channel-level signal integrity.

In this work, we focus on FOWLP-oriented implementation of fine-pitch RDLs for advanced packaging. We introduce a chemically amplified positive photoresist and systematically optimize key photolithography parameters—exposure energy, focus offset, and soft-bake conditions—to establish a manufacturable process window for reliable 2 μm /2 μm L/S fabrication. The approach emphasizes improved LER, sidewall quality, and post-plating structural robustness while minimizing dependence on overly complex

multilayer stacks or specialized tooling. The results are intended to support the practical deployment of fine-pitch RDLs in FOWLP for advanced packages, with attention given to scalability and compatibility with downstream processes [1,4].

In summary, the key contributions of this work are as follows:

- This study experimentally establishes a manufacturable process window for 2 μm /2 μm CAR-based RDL lithography in an FOWLP environment by systematically varying exposure energy, focus offset, and thermal processing conditions, and by quantifying their impact on pattern collapse and CD variation.
- This study demonstrates, through FIB cross-sectional analysis, that the optimized process yields electroplated RDL structures with stable sidewall profiles and structural integrity over relatively thick and non-planar dielectric stacks.
- This study correlates the lithography and electroplating results with channel-level HBM3-class signal integrity simulations, thereby linking fine-pitch RDL geometry and process-induced variations to jitter, noise, and impedance mismatch in high-speed operation.
- This study provides practical guidelines and process constraints for integrating 2 μm class RDLs into an FOWLP-based 2.5D interposer and embedded bridge architectures, highlighting the trade-offs between lithographic resolution, pattern robustness, and high-speed electrical performance.

2. Materials and Methods

To ensure high pattern fidelity is essential for stable electrical performance and yield in semiconductor devices, this study aims to establish optimized photolithography parameters through systematic experimentation and analysis. By evaluating the influence of various lithographic conditions, we provide valuable insights into scalable manufacturing for next-generation semiconductor packaging.

Figure 1 schematically presents the overall process flow for PR pattern formation and RDL implementation.

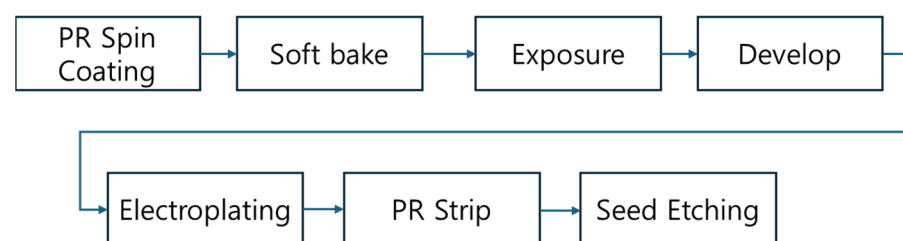


Figure 1. Typical process flow diagram for PR pattern formation and RDL.

For fine pattern formation, a CAR resist was selected due to its superior performance in micro-lithography. CAR resists generate acid upon UV exposure via a photo acid generator (PAG), which then catalyzes chemical reactions within the resist matrix during the post-exposure bake (PEB) step. This process increases the solubility of the exposed regions, enabling selective removal during development. CAR resists consist of a polymer, PAG, and solvent. The acid generated during exposure does not degrade but acts catalytically, interacting with multiple polymer chains, hence the term “chemically amplified.” This mechanism allows CAR to achieve high contrast even with short exposure times [11].

The chemical amplification characteristics of CAR enable extremely high resolution and robust fine-feature formation. In advanced packaging applications, lithographic fidelity and pattern stability are critical for reliable electrical performance. In this work, we focus on optimizing exposure and thermal processing conditions to achieve stable 2 μm /2 μm line/space patterns suitable for FOWLP-based RDL fabrication [11,12,14]. Furthermore,

the reaction front propagation model suggests that acid diffusion during PEB is enhanced by transient free volume generation and plasticization, which accelerates deprotection but may also lead to pattern distortion if uncontrolled [11]. Lithographic performance is highly sensitive to PEB temperature relative to the polymer’s glass transition temperature (Tg); optimal resolution is achieved near Tg, while excessive temperatures can cause resist hardening and linewidth narrowing [11].

2.1. Photoresist Material and Thickness Control

Photoresist (PR) materials are essential for defining fine patterns with high precision in semiconductors and advanced packaging processes. The PXi-series CAR, provided by a Japanese manufacturer, exhibits robust line/space patterning capability and is optimized for applications requiring fine pattern fidelity and film-thickness uniformity.

Table 1 provides information on the material properties and characteristics available from the manufacturer.

Table 1. The minimum material property information authorized for release by the material development company.

Viscosity	DOF	Design Film Thickness	Resolution
80 cp	25 μm	3~12 μm	L/S 2/2 μm Capability

Accurate control of PR thickness is critical because it directly affects subsequent electroplating step heights and interconnect geometry. In parallel, maintaining low resist roughness is important to preserve pattern fidelity in fine-pitch RDLs. The PXi-series CAR used in this work is designed to provide stable film thickness and pattern quality for 2 μm/2 μm line/space applications.

Variations in PR thickness can significantly impact the final product’s quality and electrical performance. Therefore, precise control of PR thickness is essential to ensure optimal outcomes in electroplating processes. In the context of fine-pitch RDL formation, this indicates that PR thickness control is not simply a geometric requirement but is a key element for completing the overall pattern structure while managing RDL thickness control in the subsequent electroplating process by controlling uniform thickness. Figure 2 illustrates the spin coating process used to control the PR thickness in this study.

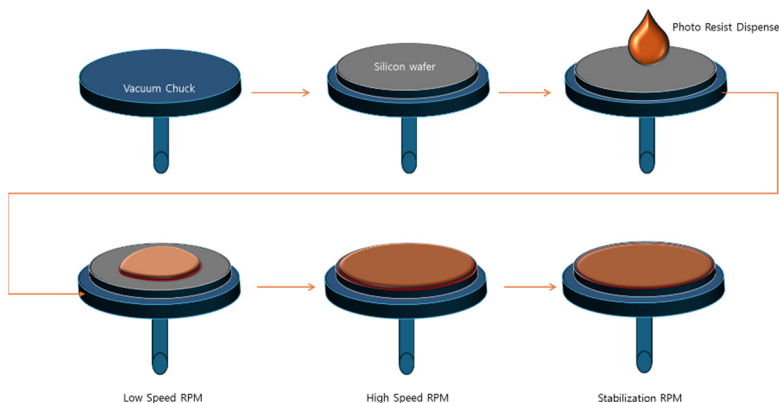


Figure 2. The spin coating process, as influenced by the viscosity of the photoresist material, using a spin coater from Brozen Co., Ltd., Seoul, Republic of Korea.

In this study, spin coating was employed to control the PR thickness. The objective was to investigate the influence of spin coating parameters—such as spin speed, solution viscosity, spin time, and dispense volume—on the resulting PR film thickness. The PR thickness was measured using an optical profiler (WL3500F, Kovis Technology Co., Ltd.,

Seoul, Republic of Korea), with measurements taken at nine different points on each sample to evaluate coating uniformity.

Spin coating is a widely adopted technique for depositing uniform thin films onto flat substrates. In this process, a small amount of liquid photoresist is dispensed at the center of the substrate, which is then rotated at a speed optimized for the material properties of the solution [15]. The centrifugal force distributes the material evenly across the surface, forming a thin film. The final film thickness is primarily influenced by the spin speed, solution viscosity, and spin duration. By precisely adjusting the spin speed according to the rheological characteristics of the photoresist, the film thickness can be accurately controlled to meet specific application requirements. Figure 1 illustrates the experimental results of controlling photoresist thickness to a target value of 5 μm using spin speeds ranging from 700 to 1000 rpm. Among the tested conditions, 800 rpm was found to be the most suitable spin speed for achieving the desired film thickness with optimal uniformity.

Figure 3 shows the image of the photoresist thickness measurement obtained using an optical profiler, following exposure and development with a fine pattern mask on a wafer coated at 800 rpm. At the selected spin speed of 800 rpm, the PR thickness was $5.0 \pm 0.5 \mu\text{m}$ across nine measurement points, corresponding to a thickness uniformity of 10%.

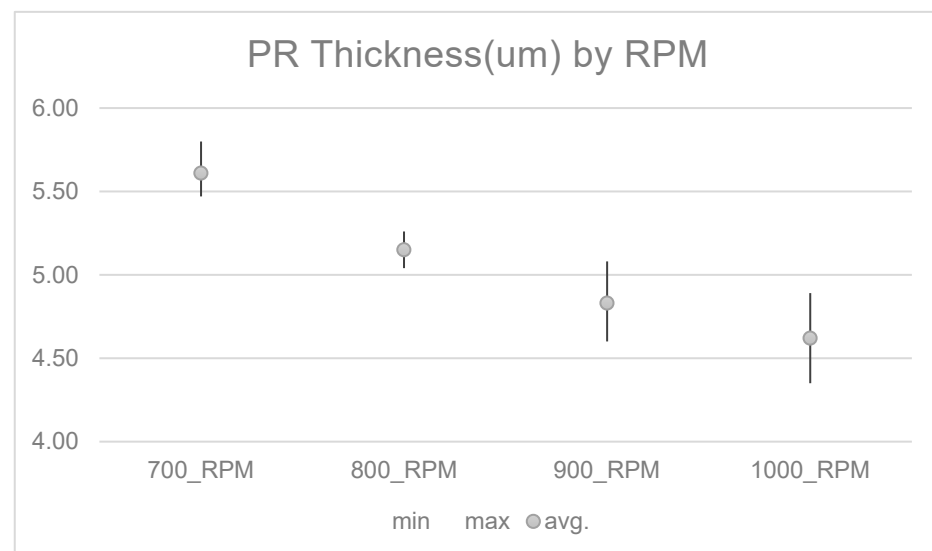


Figure 3. Coating Thickness Variation by Spin RPM.

2.2. Exposure and Soft Bake Process Conditions

To evaluate the resolution and pattern fidelity of fine-pitch photoresist structures, a series of photolithography experiments were conducted using a broadband exposure system operating in the 300–500 nm wavelength range. The exposure energy was varied from 90 mJ/cm^2 to 300 mJ/cm^2 , and the focus offset was adjusted within a $\pm 5 \mu\text{m}$ range to assess the impact on critical dimension (CD) control.

In parallel, the soft bake (prebake) process was optimized to ensure uniform solvent evaporation and stable resist film formation prior to exposure. The soft bake temperature varied between 90 $^{\circ}\text{C}$ and 130 $^{\circ}\text{C}$, and the bake time was adjusted from 60 to 180 s. These parameters were systematically correlated with the exposure conditions to determine their combined effect on resist profile, resolution, and pattern collapse resistance [13].

The experimental matrix was designed to explore the interdependence between exposure wavelength, energy dose, focus setting, and soft bake conditions. This approach enabled the identification of optimal process windows for achieving high-resolution 2 $\mu\text{m}/2 \mu\text{m}$ line/space patterns using CAR resist materials.

Based on the experimental results obtained under the aforementioned conditions, this study defined acceptable CD tolerance as $\pm 10\%$ of the target ($2\text{ }\mu\text{m}$) and evaluated pattern collapse probability as failure criteria. The optimal conditions were clearly distinguished and annotated for each parameter set. Figure 4 was obtained by separately evaluating region-specific exposure conditions within a single wafer and the effects of different soft bake temperatures across different wafers. As illustrated in Figure 5a, soft bake at $90\text{ }^{\circ}\text{C}$ led to incomplete solvent removal, resulting in locally swollen resist features and degraded pattern definition in some areas. In contrast, the $120\text{ }^{\circ}\text{C}$ soft bake condition in Figure 5b produced a more uniform resist film, but excessively elevated temperatures caused line narrowing and profile deformation due to over-hardening of the CAR resist. Figure 5 shows microscope images of the patterns after formation. Figure 6 presents a graph of the measured pattern space (a) and line width (b) for each condition.

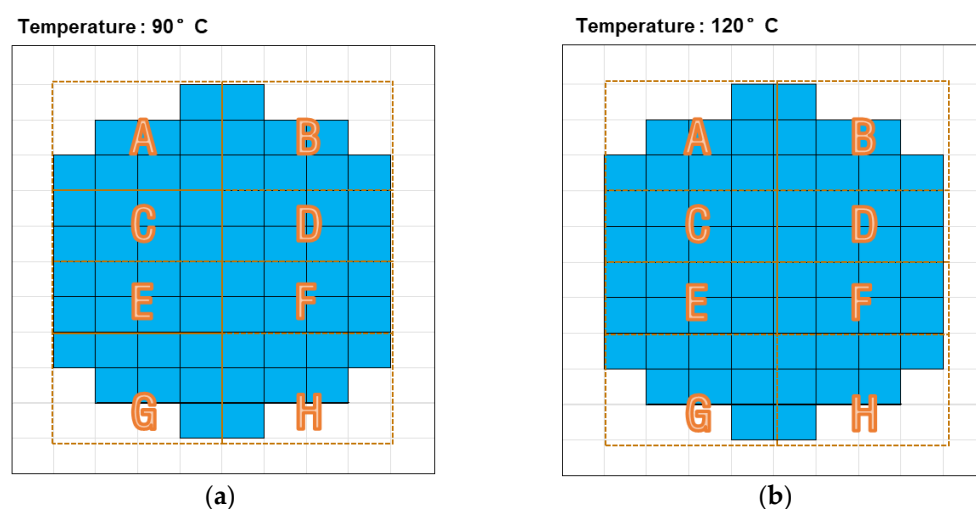


Figure 4. (a) Exposure process conducted on a wafer with soft bake conditions set at $90\text{ }^{\circ}\text{C}$; (b) soft bake conditions set at $120\text{ }^{\circ}\text{C}$. For both wafers, exposure conditions were categorized from A to H based on different positional parameters across the wafer surface. The solid lines indicate the package area within the wafer, while the dashed lines represent the exposure condition regions that were separately applied for each condition.

Under the optimized exposure and focus conditions, the measured Pattern width and space were $2.0 \pm 0.5\text{ }\mu\text{m}$ and $2.0 \pm 0.5\text{ }\mu\text{m}$, respectively ($n = 8$), which satisfy the $\pm 25\%$ CD tolerance criterion.

Figure 7 shows the image obtained from measurements of photoresist thickness and fine-pattern regions after exposure and development. The wafer was coated with photoresist at 800 rpm and exposed using a fine-pattern mask with $2\text{ }\mu\text{m}/2\text{ }\mu\text{m}$ line/space features. The exposure energy and development conditions were optimized for the given process, and the measurements were performed using a thickness measurement WL3500F (thickness resolution 10 nm) model from Kovi Technology Co., Ltd. Due to the fixed scanning orientation of the optical profiler, the measurement cross-section in Figure 7b appears slightly tilted with respect to the wafer's nominal Y-axis. However, the measurement points themselves are located on a regular grid, and the tilt does not affect the extracted thickness values or the evaluated uniformity.

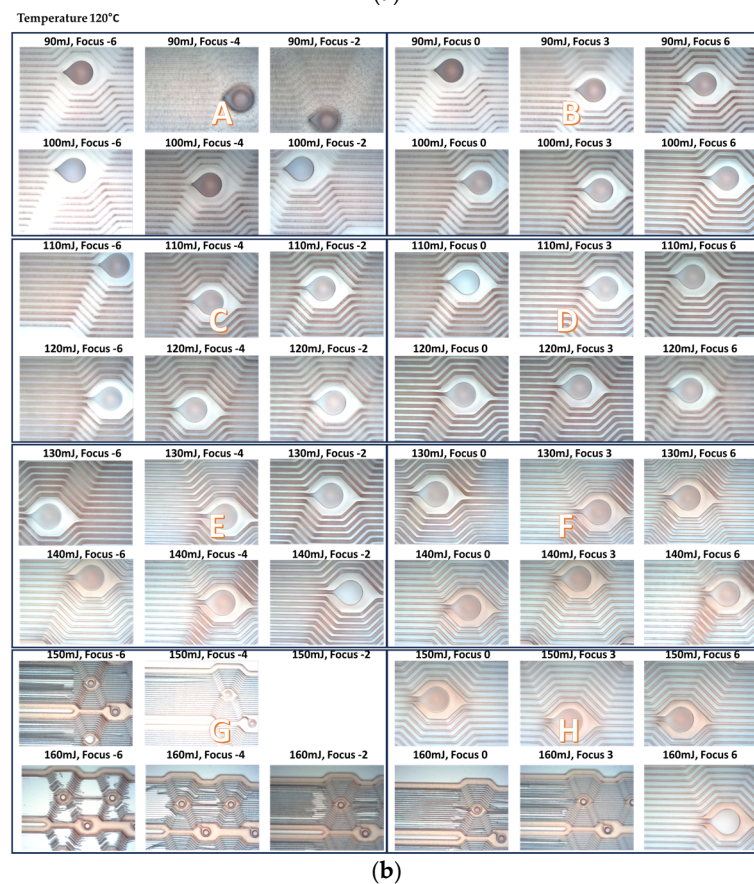
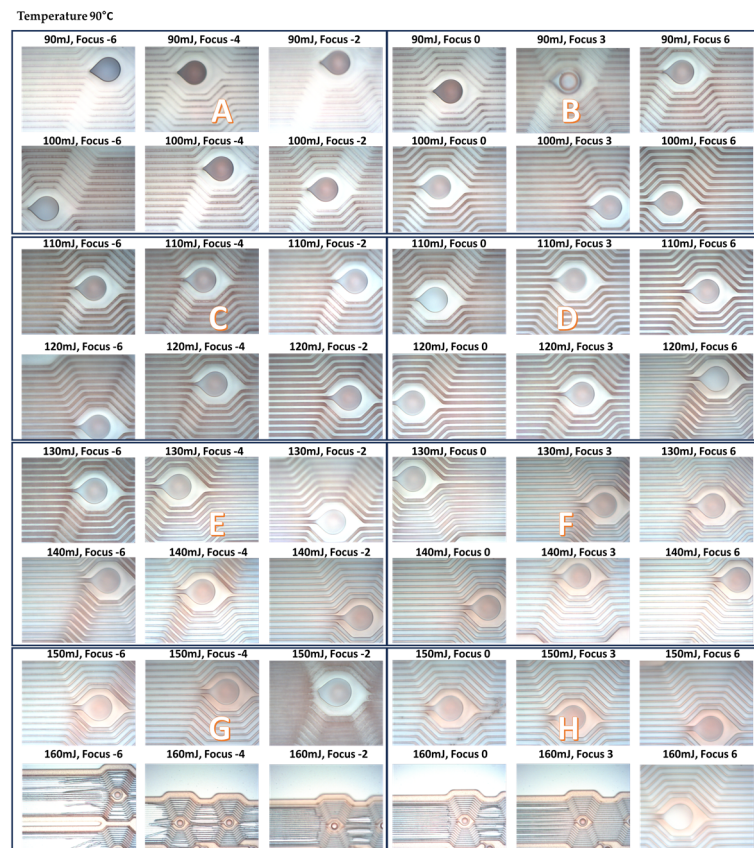


Figure 5. (a) Exposure process conducted on a wafer with soft bake conditions set at 90 °C; (b) exposure process conducted on a wafer with soft bake conditions set at 120 °C. For both wafers, exposure conditions were categorized from A to H based on different positional parameters across the wafer surface.



Figure 6. This graph shows the measured results of pattern space (a) and line width (b) as a function of exposure conditions and focus variation.

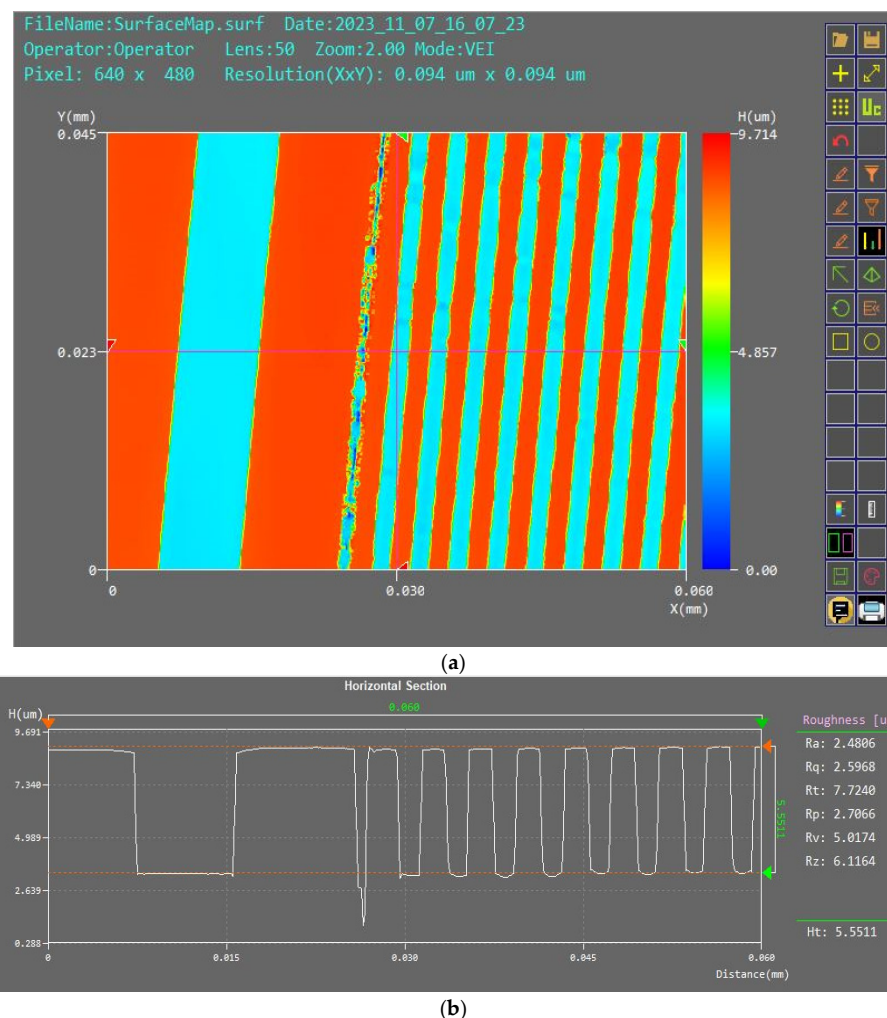


Figure 7. (a) The screen image of the completed 2 $\mu\text{m}/2 \mu\text{m}$ line/space fine pattern region after the photolithography process. (b) The corresponding screen image showing the measured photoresist thickness profile across the patterned area.

Figure 8 presents the cross-sectional analysis performed to verify the final structure of the RDL after completing the photolithography and electroplating processes, followed

by the photoresist stripping step and prior to seed layer etching. To clearly observe the formation of the $2\ \mu\text{m}/2\ \mu\text{m}$ line/space structure, a Focused Ion Beam (FIB) image was acquired using an EsB Model from Zeiss Co., Ltd., Seoul, Republic of Korea.

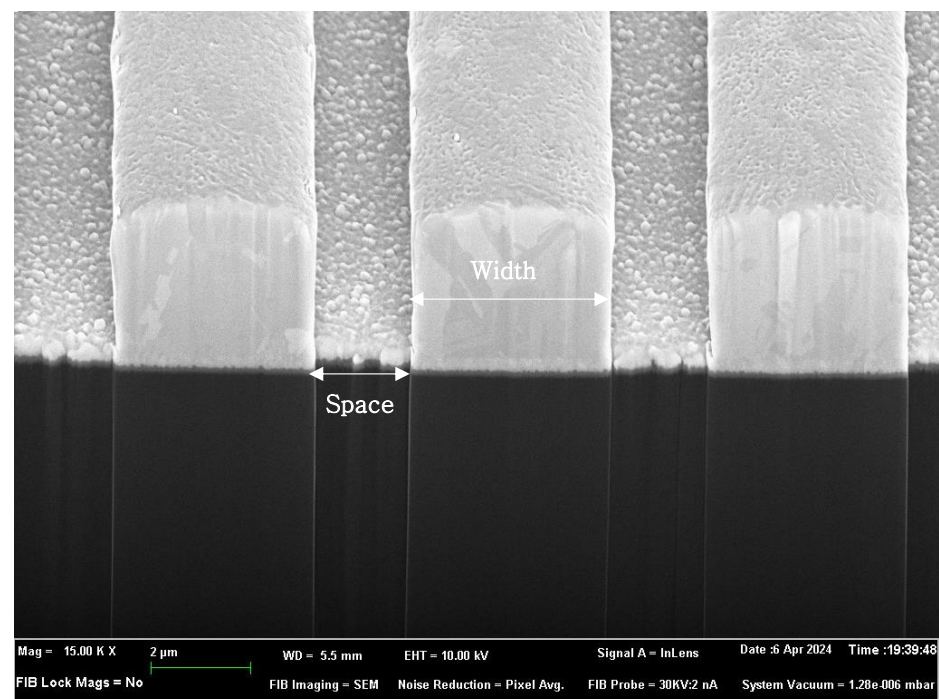


Figure 8. FIB cross-sectional image of the L/S $2\ \mu\text{m}/2\ \mu\text{m}$ region, obtained after photoresist patterning, electroplating, and completion of the PR strip process, used to verify that the $2\ \mu\text{m}/2\ \mu\text{m}$ photo pattern has been correctly formed.

As shown in Figure 9, since the seed etching process has not yet been completed, the measured RDL width appears larger compared to the space. This will be modified by the etch ratio during the etching process. Therefore, it is necessary to review the measurement results after etching. Quantitatively, the average measured RDL width and space before seed etching were $2.7\ \mu\text{m}$ and $1.1\ \mu\text{m}$, respectively, compared to the target values of $2.0\ \mu\text{m}/2.0\ \mu\text{m}$. When considering the sum of width and space, the measured pitch remained close to the intended $4.0\ \mu\text{m}$, indicating that the overall routing density is preserved even before final seed etching.

As shown in Figure 10, after etching, the average result was confirmed as Width 2.0 , deviation $0.3\ \mu\text{m}$, and Space $1.8\ \mu\text{m}$, deviation $0.3\ \mu\text{m}$. The measured RDL width and space were $2.0 \pm 0.5\ \mu\text{m}$ and $2.0 \pm 0.5\ \mu\text{m}$, respectively ($n = 8$), which satisfy the $\pm 25\%$ CD tolerance criterion.

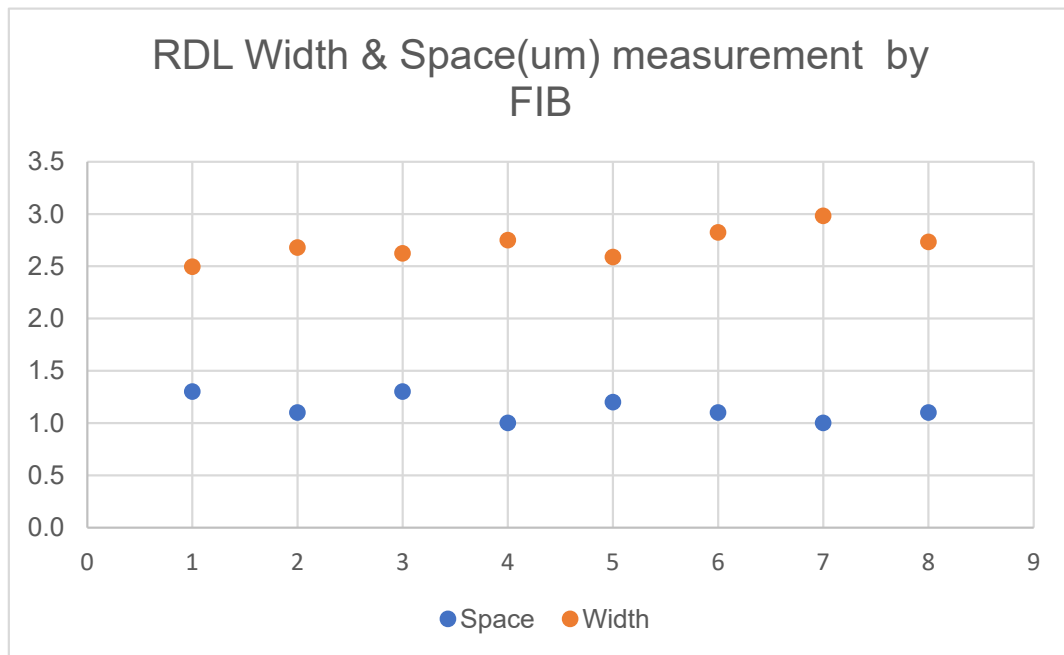


Figure 9. FIB cross-sectional analysis results of RDL line width and spacing measurements.

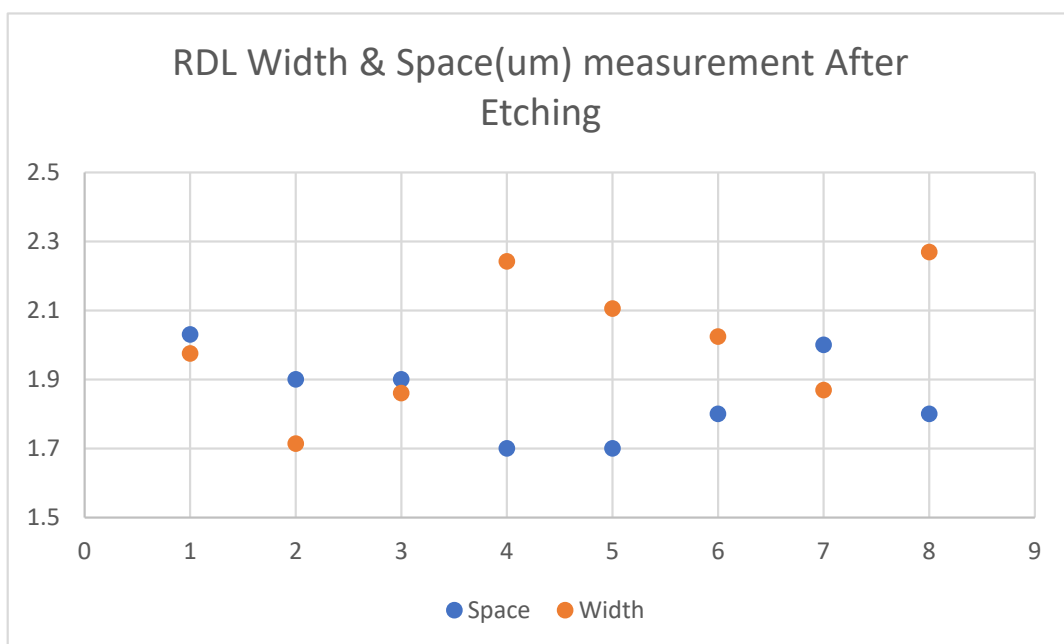


Figure 10. Results of RDL line width and spacing measurements after Etching.

Figure 11 images were obtained after completing the electroplating and etching processes using patterns formed under various exposure conditions. The results confirm that exposure conditions had no significant impact within the optimized process window on the final pattern formation, as no observable differences were detected in the structural integrity or geometry of the plated features.

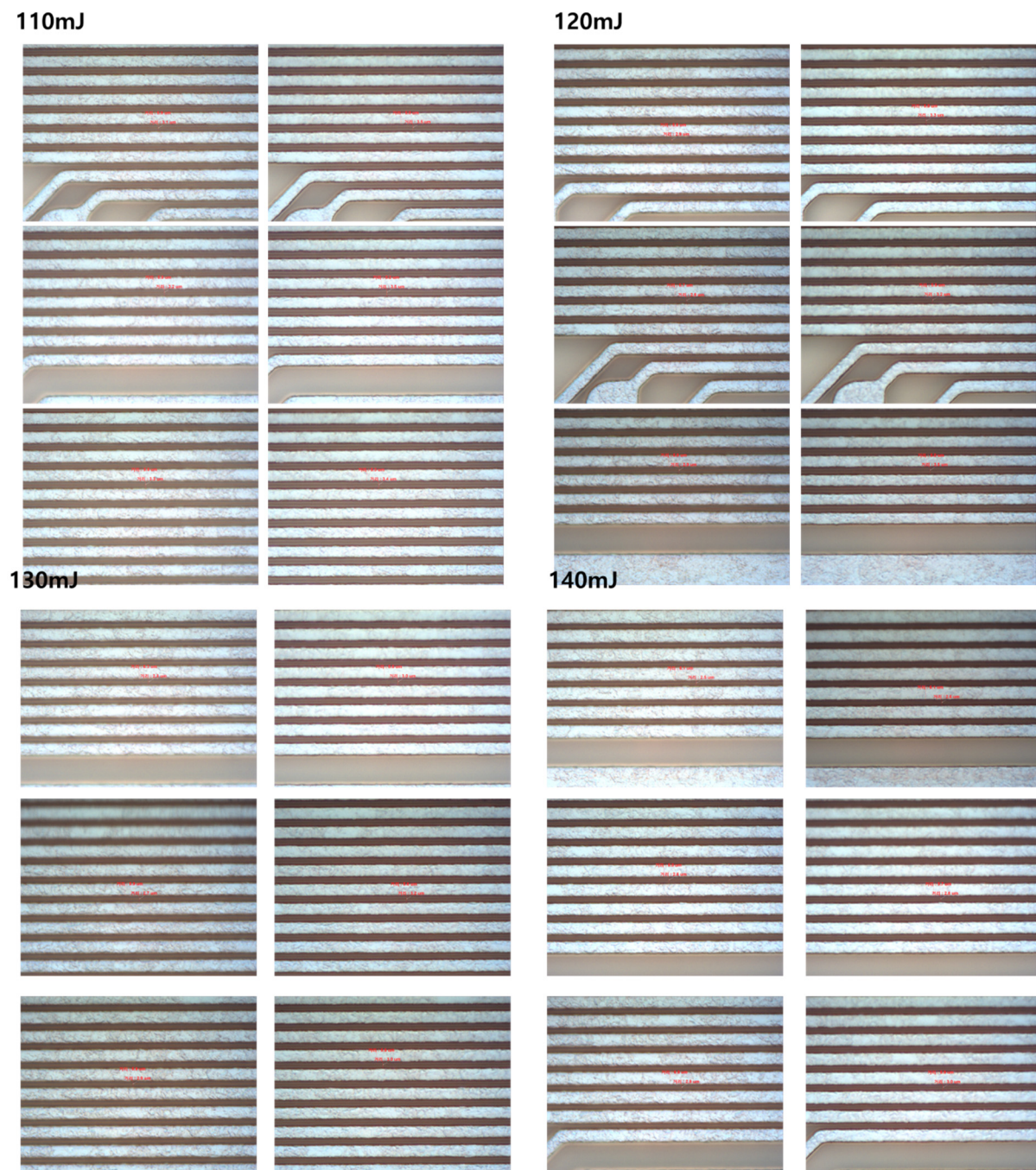


Figure 11. This image shows pattern formation under different exposure energy conditions and the verification of RDL formation by measuring the structure after Cu electroplating, PR stripping, and seed metal etching.

3. Results

This study presents an optimized photolithography process for fine-pitch RDL fabrication in a FOWLP environment. The process was assessed in terms of pattern fidelity, dimensional stability, and manufacturability for $2\ \mu\text{m}/2\ \mu\text{m}$ line/space (L/S) RDL structures. Through systematic variation in exposure energy and focus conditions, stable fine-pitch RDL patterns were reproducibly obtained, addressing process limitations commonly encountered in sub- $5\ \mu\text{m}$ RDL fabrication [1].

A comprehensive analysis was performed to investigate the effects of key lithographic parameters, including exposure energy, focus offset, soft bake temperature, and bake time, on fine-pattern formation. These parameters were found to strongly influence the structural

integrity and dimensional accuracy of the L/S features. In particular, the soft bake process primarily contributed to solvent removal and resist film uniformity, whereas the PEB conditions governed acid diffusion and resolution behavior in the chemically amplified resist system [11].

At insufficient exposure energies, incomplete resist development resulted in pattern collapse and deformation, which is consistent with capillary-force-induced collapse mechanisms and reduced mechanical stability of underexposed resist features [11,16]. Conversely, excessive soft bake temperatures caused narrowing of the patterned lines, indicating resist over-hardening associated with accelerated crosslinking and suppressed acid mobility [11]. These observations highlight the importance of balancing optical and thermal process parameters to maintain both resolution and pattern stability.

Based on these results, an optimized process window was identified that enabled uniform formation of 2 μm /2 μm L/S RDL patterns with high structural fidelity. The optimized resist thickness and pattern geometry were verified through FIB cross-sectional analysis after electroplating and prior to seed layer etching, confirming the integrity of the fabricated RDL structures.

To further assess the functional feasibility of the fabricated fine-pitch RDLs, electrical signal integrity was evaluated through channel-level simulations. Frequency-domain electrical analysis was performed using JEDEC-defined HBM3 pin assignments, and twelve representative micro-strip signal nets were selected from a sample HBM3 channel (Figure 12). The resulting simulated eye diagram indicates that the fine-pitch RDL interconnects are capable of supporting stable signal transmission.

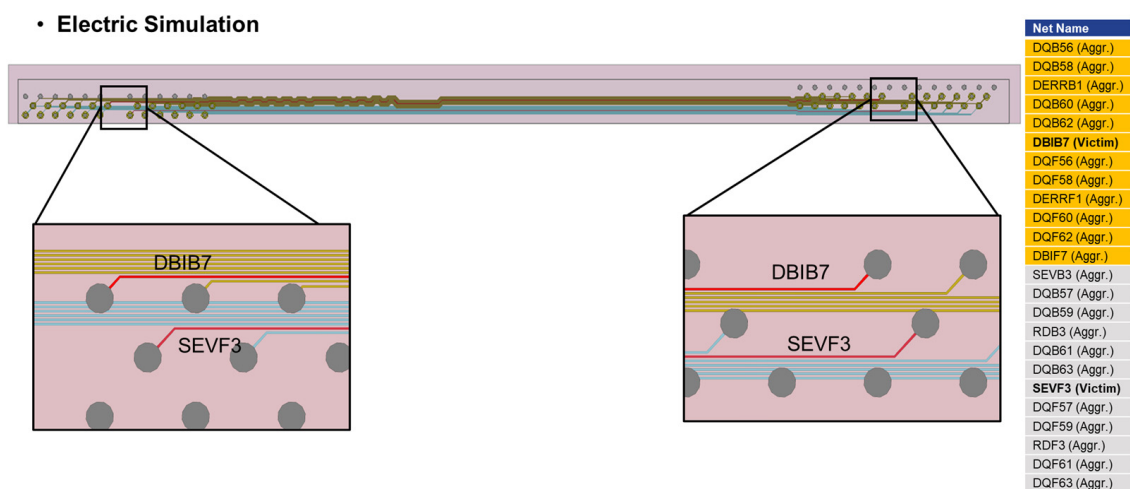
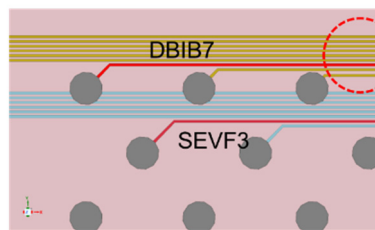


Figure 12. Eye diagram obtained from frequency-domain electrical simulation of an HBM3 channel implemented on a 2 μm /2 μm line/space RDL. Twelve micro-strip signal nets were selected from a representative HBM3 channel based on JEDEC-defined pin assignments (highlighted in yellow), and the simulated eye diagram demonstrates stable signal transmission through the fine-pitch RDL.

A more detailed channel simulation was conducted by designating the centrally located DBIB7 net as the victim signal and assigning neighboring nets as aggressors to evaluate crosstalk effects (Figure 13). The simulation was based on a Touchstone file extracted from frequency-domain analysis, with pseudo-random bit sequence (PRBS) excitation applied to each driver to approximate HBM3 signaling behavior, given the absence of a publicly available HBM3 IBIS model.

• Source Information



Name	Value
vlow	-0.3V
vhigh	0.8V
TR	21.42875ps
TF	21.42875ps
BPS	6.4Gbps
Internal Resistance	50ohm

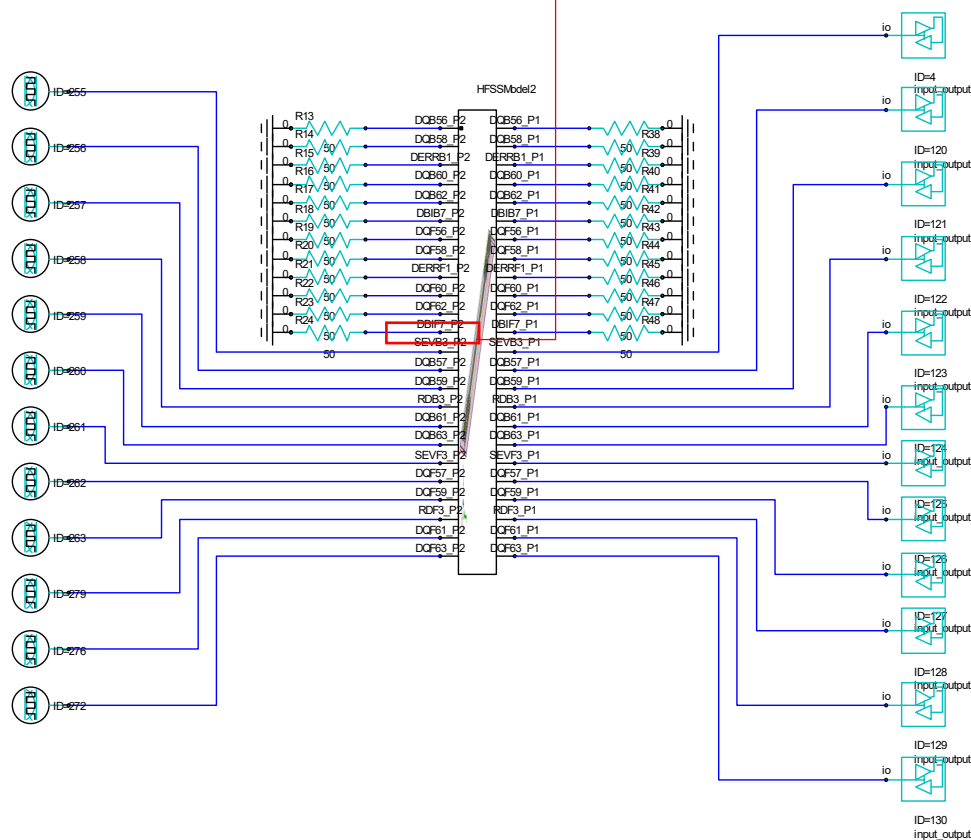


Figure 13. Channel eye diagram simulation of an HBM3 signal routed through the 2 $\mu\text{m}/2 \mu\text{m}$ RDL. A Touchstone file was generated from frequency-domain electrical analysis, and the centrally located DBIB7 signal was designated as the victim, while the remaining neighboring signals were configured as aggressors. PRBS input signals approximating HBM3 operating conditions were applied to each driver due to the unavailability of a vendor-provided HBM3 IBIS model.

Figure 14 presents the simulated eye diagram overlaid with the JEDEC standard eye mask. While the eye opening indicates a moderate level of signal integrity, the presence of jitter and noise suggests that further optimization is required. These results indicate that precise impedance matching of fine-pitch RDL transmission lines will be an important factor for improving high-speed signal performance in future process refinements.

• 6.4Gbps Eye Diagram (DBIB7)

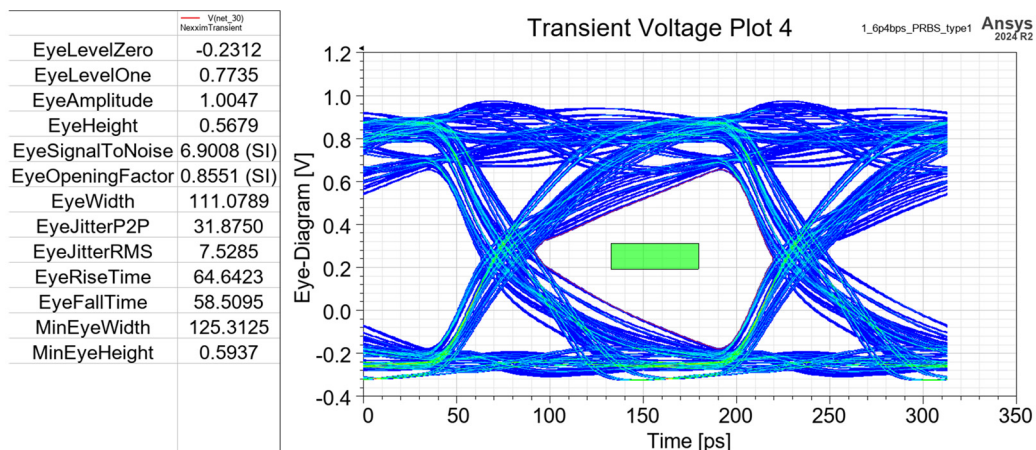


Figure 14. Simulated eye diagram overlaid with the JEDEC standard eye mask for the analyzed HBM3-class RDL channel. The eye opening meets an average signal integrity level, while noticeable jitter and noise are observed. Taken together, these lithographic and structural results establish a direct link between the defined process window and the manufacturable realization of 2 μm /2 μm RDLs suitable for high-speed signaling, rather than merely demonstrating isolated patterning capability.

4. Discussion

The results of this study demonstrate that a CAR-based photolithography process can be effectively applied to the fabrication of 2 μm /2 μm fine-pitch RDL structures in a FOWLP environment. By systematically optimizing exposure energy, focus offset, and thermal processing conditions, a stable process window was established that enables reproducible formation of fine-pitch RDLs with acceptable structural fidelity.

The observed dependencies of pattern quality on exposure and baking conditions are consistent with known behavior of chemically amplified resists. Insufficient exposure resulted in incomplete deprotection and reduced mechanical robustness, leading to pattern collapse, whereas excessive soft bake temperatures promoted resist hardening and limited acid diffusion, causing linewidth narrowing. These trends highlight the need for balanced optical and thermal control, particularly when targeting sub-5 μm RDL geometries on relatively thick dielectric stacks typical of FOWLP processes.

4.1. Mechanistic Interpretation of CAR-Based Fine-Pitch RDL Lithography

The sensitivity of pattern quality to exposure energy and baking conditions can be interpreted in terms of acid generation, diffusion, and polymer deprotection in chemically amplified resists. At insufficient exposure energies, the amount of photoacid generated by the PAG is inadequate to fully deprotect the resist polymer matrix, resulting in partially weakened features that are prone to capillary-force-induced collapse during the development and rinse steps [11,16]. This explains the experimentally observed pattern collapse and deformation under low-dose conditions.

In contrast, excessively high soft bake or post-exposure bake temperatures reduce resist viscosity and increase free volume, which accelerates acid diffusion and deprotection reactions [11]. While enhanced acid diffusion can improve line-edge roughness, it also broadens the reaction front, causing linewidth variation and profile distortion for 2 μm /2 μm features. Moreover, over-baking promotes resist hardening and suppresses subsequent acid mobility, which leads to linewidth narrowing and degraded CD control, as observed in our experiments.

The downstream electroplating and resist stripping steps partially compensate for lithography-induced CD variations. Because the effective RDL width is determined not

only by the resist opening but also by the local current density and plating kinetics within the trench, slight differences in resist profile tend to be averaged out during metal growth. Nevertheless, this self-compensation effect is limited and does not remove the need for a tightly controlled lithography process window, especially when targeting sub-5 μm RDLs over thick dielectric stacks in FOWLP.

The FIB cross-section measurement results were measured in advance to verify the pattern of the photoresist before etching, and the post-etch measurement results confirmed that the error range ($\pm 25\%$) was within an acceptable level for the 2 μm /2 μm design rule in the FOWLP-based interposer.

Although the lithographic optimization strongly influenced resist profile and pattern fidelity, the subsequent electroplating and resist stripping steps were found to partially mitigate variations introduced during exposure. This observation suggests that downstream metallization processes can relax certain lithographic sensitivities; however, they do not eliminate the need for a well-defined lithography process window, especially for ensuring repeatability and manufacturability.

Electrical signal integrity was evaluated through channel-level simulations to complement the structural analysis. The simulated eye diagrams indicate that the fabricated 2 μm /2 μm RDL structures can support stable signal transmission under HBM3-class signaling conditions. At the same time, the presence of jitter and noise observed in the eye mask analysis indicates that the current RDL geometry does not fully optimize impedance control. These results emphasize that lithographic resolution alone is not sufficient to guarantee high-speed electrical performance, and that coordinated optimization of RDL geometry, dielectric properties, and routing design is required.

It should be noted that the electrical analysis presented in this work is based on frequency-domain simulation and PRBS-based excitation in the absence of a vendor-provided HBM3 IBIS model. As such, the simulation results are intended to provide qualitative insight into signal integrity trends rather than full compliance verification. Future work will focus on incorporating more accurate transmitter and receiver models, as well as experimental electrical measurements, to further validate the proposed RDL process.

4.2. Application Perspective and Future Extension

The demonstrated 2 μm /2 μm RDL process directly targets the interconnect density and routing constraints of advanced 2.5D interposer applications, particularly for HBM3 integration and chiplet-based AI/HPC systems [1,4,5]. The channel-level simulations confirm that the fabricated fine-pitch RDLs can sustain HBM3-class signaling with a moderate level of signal integrity, indicating their suitability as a baseline technology for high-bandwidth memory interfaces and short-reach die-to-die links.

However, the observed jitter and noise in the simulated eye diagrams also reveal that the current RDL geometry does not yet provide fully optimized impedance matching or crosstalk suppression for strict JEDEC-compliant operation. To translate the proposed process into production-ready interposers and embedded bridge solutions, further work is required on:

- (i) Multilayer RDL stacks with tighter impedance control and shielding structures;
- (ii) Co-optimization of RDL geometry and dielectric properties for improved signal integrity;
- (iii) Experimental validation of electrical and reliability performance under realistic system conditions.

5. Conclusions

This study investigated the fabrication of fine-pitch redistribution layers with 2 $\mu\text{m}/2\ \mu\text{m}$ line/space geometry using a CAR-based photolithography process optimized for fine-pitch RDL fabrication in a fan-out wafer-level packaging environment. Through systematic optimization of exposure and thermal processing conditions, a process window suitable for reproducible fabrication was established, enabling reproducible formation of fine-pitch RDL structures with stable pattern fidelity. Furthermore, the realization of 2 $\mu\text{m}/2\ \mu\text{m}$ RDL technology is required to meet the interconnect density and routing constraints of advanced 2.5D interposer applications [5].

Structural validation through cross-sectional analysis confirmed the integrity of the electroplated RDL features, while channel-level electrical simulations demonstrated that the fabricated RDL interconnects are capable of supporting HBM3-class signal transmission with a moderate level of signal integrity. The presence of jitter and noise observed in the simulated eye diagrams indicates that further optimization, particularly in impedance matching of the fine-pitch RDL transmission lines, is required to improve high-speed performance.

Rather than presenting a fully optimized interposer solution, this work provides an engineering-level assessment of the lithographic and process constraints associated with implementing 2 μm RDLs in FOWLP-based interposers. The results offer practical insight into process window definition and highlight key considerations for integrating fine-pitch RDLs into advanced packaging platforms. Future studies will focus on multi-layer RDL implementation, improved impedance control, and experimental electrical and reliability validation to further extend the applicability of the proposed process.

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Conflicts of Interest: Authors Jung Won Lee, Sunghyuk Lee, Jay Kim, Lewis Kang, Han Ju Yu, Min Ji Lee, Seong Hwan Han, Jae Kyung Lee, Hailey Hwang, Jung Gi Kim, Chang Young Hong, Jade Park, Su Hyun Kim and Myeung Jin Kim were employed by the company Nepes. The remaining author declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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